Technical Document #1030: Cybersecurity - Comprehensive Overview

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Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Zero trust architecture assumes no implicit trust and verifies every request.
- Penetration testing simulates cyber attacks to identify vulnerabilities.
- Security information and event management (SIEM) systems collect and analyze security logs.